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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/930,759	09/09/2022	Mariusz Bulawa	92028_97306_0015_GJG	1933

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EXAMINER	
LIN, CHENG XI	

ART UNIT	PAPER NUMBER
3615	

MAIL DATE	DELIVERY MODE
06/23/2025	PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Office Action Summary	Application No. 17/930,759	Applicant(s) Bulawa et al.	
	Examiner CHENG X LIN	Art Unit 3615	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 09 September 2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3,7-8 and 10-13 is/are rejected.

8) ☑ Claim(s) 2,4-6,9 and 14-20 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 09 September 2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☑ Other: See Continuation Sheet.

Continuation of Attachment(s) 4) Other: EP 1808531 A1, EP 3567161 A1, CN109398423A, CN 110316225 A, DE 102017103243 B3

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

2. This is the first non-final office action on the merits. Claims 1-20 are currently pending.

Priority

3. Acknowledgment is made of applicant's claim for foreign priority under 35 U.S.C. 119 (a)-(d). The certified copy has been filed in parent Application No. PLP.438929, filed on 09/10/2021.

Information Disclosure Statement

4. The information disclosure statements (IDS) submitted on 10/07/2022 and 03/31/2023 have been received and considered by the examiner.

Drawings

5. The drawings are accepted.

Claim Objections

6. Claim 3 is objected to because of the following informalities: Claim 3 recites the phrase "a one of the connection elements". It appears this should read "one of the connection elements". Appropriate correction is required.

Claim Rejections - 35 USC § 103

7. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. Claim(s) 1, 3, 7-8, and 10-13 is/are rejected under 35 U.S.C. 103 as being unpatentable over Buchinger et al. (US 20210347395 A1, known hereinafter as Buchinger '395), in view of Buchinger (EP 1808531 A1, known hereinafter as Buchinger '531).

10. Regarding claim 1, Buchinger '395 teaches (Fig. 1-5): A track sensor arrangement (sensor arrangement 20) consisting of a wheel sensor (wheel sensor 21), a detachable connection assembly (connector 23) and a track clamp (carrier 22 and clamp part 34), wherein the detachable connection assembly (23) consists of a supply cable (power cable 33) and a plug (connector 23) provided with an electric plug connection (plug 31 having electrical contacts 24)(Fig. 5), whereas the track clamp (22, 34) consists of a support (carrier 22) and of connection elements (22, 34)(Fig. 1A-1B), wherein inside the body of the plug (23) an assembly compartment (cylinder shaped part 40 for plug 31; para. 0045, lines 11-13) is formed (Fig. 4-5), and the body of the plug (23) is slidingly fitted (slidable in the up-down direction within elongated hole 30 of the second mounting part 28; para. 0043, lines 4-7) on the support (carrier 22)(Fig. 2-3).

Buchinger '395 further teaches (Fig. 1C, 2, and 5): a mounting holder (annotated Fig. 5 below) fastened to the support (carrier 22) by means of a screw connection (screws 26; para. 0041; Fig. 1C and 2), but does not explicitly teach that the mounting holder is slidingly fitted on the body of the plug.

However, Buchinger '531 teaches an alternate track sensor arrangement, wherein (Fig. 1 and 3): a mounting holder (cover 17) is slidingly fitted on a body of a plug (plug 15) and fastened to a support (housing 3) by means of a screw connection (screws 19)(para. 0014, lines 159-162).

It would have been obvious to a person of ordinary skill in the art, before the effective filing date of the claimed invention, for Buchinger '395 to slidingly fit the mounting holder on the body of the plug, as disclosed by Buchinger '531, with a reasonable expectation of success because it would enable the mounting portion to be independently manufactured and be removably attached to the body of the plug for ease of assembly or replacement.

11. Regarding claim 3, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): the support (22) is used as one of the connection elements (22, 34) in the track clamp (22, 34)(Fig. 2).

12. Regarding claim 7, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): the mounting holder (annotated Fig. 5 below), but does not teach that a narrowing is formed between wings of the mounting holder.

However, Buchinger '531 teaches an alternate track sensor arrangement, wherein (Fig. 1 and 3): a mounting holder (cover 17) has a narrowing (downwardly open insertion opening 18 with tapered surface) formed between wings (22) of the mounting holder (17)(Fig. 3).

It would have been obvious to a person of ordinary skill in the art, before the effective filing date of the claimed invention, for Buchinger '395 to have a narrowing formed between wings of the mounting holder, as disclosed by Buchinger '531, with a reasonable expectation of success because it

would enable the mounting portion to be shaped and adapted to fit the body of the plug for a secured attachment.

13. Regarding claim 8, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): in the detachable connection assembly (23) a through hole (holes 30) is formed (Fig. 5), and in the support (22) a concentric recess (two circular holes 30 on second mounting part 28) is formed (Fig. 3), in which a seal or an access authorization screw (screw 26) is embedded (Fig. 1C and 2).

14. Regarding claim 10, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): in the body of the plug (23) a socket (plug 31) is formed, in which an elastic protective cover (sealings rings or gaskets 41) of the electric plug connection (24, 31) is situated.

15. Regarding claim 11, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): a roughened manipulation area (see backside of carrier 22; Fig. 3) is formed on the support (22).

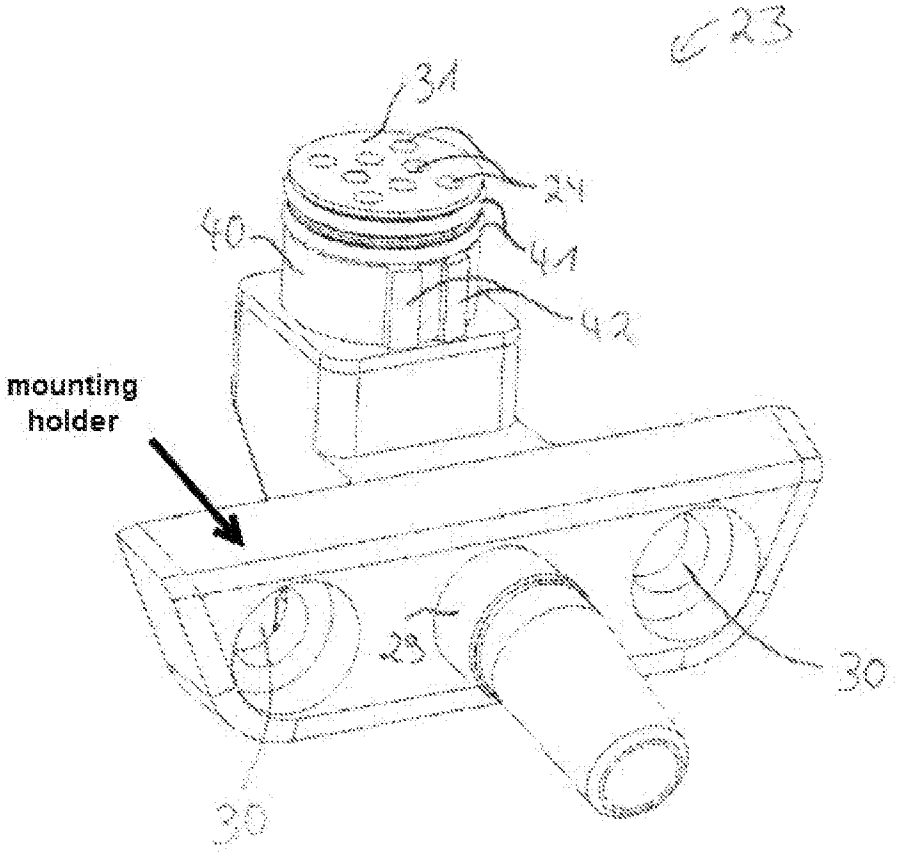
16. Regarding claim 12, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): the track clamp (22, 34) is intended for attaching the sensor arrangement to a rail (Fig. 1A-1B).

17. Regarding claim 13, Buchinger '395 and Buchinger '531 teach the elements of claim 1, as stated above. Buchinger '395 further teaches (Fig. 1-5): the track clamp (22, 34) is intended for attaching the sensor arrangement to a rail (Fig. 1A-1B), but does not teach that the track clamp (22, 34) is intended for attaching the sensor arrangement to a railway sleeper.

However, it is known to one of ordinary skill in the art that a rail is commonly attached to a railway sleeper. Therefore, it would have been obvious to a person of ordinary skill in the art, before the effective filing date of the claimed invention, for Buchinger '395 to attach the track clamp to the railway

sleeper through the attachment to a rail, with a reasonable expectation of success because it would ensure that the sensor arrangement is stable and supported on a railway sleeper on a ballast.

FIG 5



Allowable Subject Matter

18. **Claims 2, 4-6, 9, and 14-20 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claims and if all claim objections and rejections in view of 35 USC § 112 second paragraph are overcome**.

The following is a statement of reasons for the indication of allowable subject matter:

19. Regarding claim 2 and its depending claim(s) 19-20, the prior art fails to teach that the mounting holder is seated above first guides which are formed on the body of the plug, wherein the first guides are seated in second guides which are formed on the support. While Buchinger '395 further teaches (Fig. 1-5): the mounting holder (annotated Fig. 5 below) is seated on the body of the plug (Fig. 5), the examiner finds no obvious reason to modify Buchinger '395 such that the body of the plug has first guides for seating the mounting holder, wherein the first guides are seated in second guides which are formed on the support. Such a modification would require improper hindsight reasoning.

It is noted that while the secondary reference Buchinger '531 teaches an alternate track sensor arrangement, wherein (Fig. 1 and 3): a mounting holder (cover 17) is slidingly fitted on a body of a plug (plug 15) and fastened to a support (housing 3) by means of a screw connection (screws 19)(para. 0014, lines 159-162), Buchinger '531 also does not teach the body of the plug having first guides for seating the mounting holder, wherein the first guides are seated in second guides which are formed on the support. Modifying Buchinger '531 would require improper hindsight reasoning and additional modifications to a modifying reference.

20. Regarding claim 4 and its depending claim(s) 14-18, the prior art fails to teach that the body of the plug is fitted on pin guides which are fixed on the support, and in through guides which are formed in the body of the plug. While Buchinger '395 further teaches (Fig. 1-5): the body of the plug (23) is fixed on the support (22) in through guides (30) which are formed in the body of the plug (23)(Fig. 5), the examiner finds no obvious reason to modify Buchinger '395 such that pin guides are also fixed on the support for fitting the body of the plug. Such a modification would require improper hindsight reasoning.

21. Regarding claim 5, the prior art fails to teach that the mounting holder is connected to the body of the plug by means of a double- sided snap-fit connection. While Buchinger '395 further teaches (Fig. 1-5): the mounting holder (annotated Fig. 5 below) is connected to the body of the plug (23)(Fig. 5), the

examiner finds no obvious reason to modify the mounting plug (annotated Fig. 5 below) to have a double-sided snap-fit connection with the body of the plug. Such a modification would require improper hindsight reasoning.

It is noted that while the secondary reference Buchinger '531 teaches an alternate track sensor arrangement, wherein (Fig. 1 and 3): a mounting holder (cover 17) is slidingly fitted on a body of a plug (plug 15) and fastened to a support (housing 3) by means of a screw connection (screws 19)(para. 0014, lines 159-162), Buchinger '531 also does not teach the mounting plug has a double-sided snap-fit connection with the body of the plug. Modifying Buchinger '531 would require improper hindsight reasoning and additional modifications to a modifying reference.

22. Regarding claim 6, the prior art fails to teach that at least one elastic element is arranged between the body of the plug and the mounting holder. While Buchinger '395 further teaches (Fig. 1-5): the body of the plug (23) and the mounting holder (annotated Fig. 5 below), the examiner finds no obvious reason to modify Bochinger '395 such that that at least one elastic element is arranged between the body of the plug and the mounting holder. Such a modification would require improper hindsight reasoning.

While another reference Kumar et al. (US 7216558 B2) teaches an alternate track sensor arrangement, wherein (Fig. 1 and 6): at least one elastic element (shock-absorbing cushion 28 made of an elastomeric material) is arranged between a shield (15) and a mounting bracket (16)(col. 4, lines 43-49; Fig. 6), Kumar also does not teach that the elastic element is arranged between a body of a plug and a mounting holder. Modifying the elastic element of Kumar into Buchinger '395 such that it is sandwiched between two components not present in Kumar would require improper hindsight reasoning.

23. Regarding claim 9, the prior art fails to teach that on the mounting holder a tag field is delimited, on which a RFID identifier is mounted. While Buchinger '395 further teaches (Fig. 1-5): a

mounting holder (11), the examiner finds no obvious reason to modify Buchinger '395 such that the mounting holder has a delimited tag field, on which a RFID identifier is mounted. Such a modification would require improper hindsight reasoning.

While another reference Ma (CN109398423A, provided with translation) teaches an alternate wheel sensing arrangement, wherein (Fig. 3): a mounting holder (antenna bracket 9) has a tag field (antenna 4), the examiner finds no obvious reason to correlate Ma's antenna bracket to the mounting holder of the primary reference Buchinger '395. Modifying the mounting holder of Buchinger '395 with the tag field of Ma's antenna bracket to be delimited would require improper hindsight reasoning.

Conclusion

24. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure:

- US-5462244-A: Teaches a system for detecting one or more vehicles, such as a train, on a rail track, includes at least one optical conductor extending near and parallel to the rail track with a light source and light detector coupled thereto. One or more sensors are coupled to the rail track and include the light conductor, which sensors affect the light attenuation in the light conductor locally upon the presence of the vehicle.
- US-5628479-A: Teaches a vital wheel detector for railways injects an oscillating electrical current into a short rail segment and produces a monitor signal in response the magnetic field around the rail.
- US-7216558-B2: Teaches a rail-mounted, wheel-sensing system is shown having a wheel sensor and a rail mounting platform. The system includes a rugged steel rail clamp including two blocks mounted on the base of the rail by a pair of bolt extending under the rail. One block is connected to a vertically adjustable mounting bracket for carrying a sensor or other device. The

mounting bracket includes a platform for the sensor. Two shields on the platform protecting a sensor mounted between them. An adapter plate may be added between the mounting bracket and the clamp for installation on lighter or heavier rails.

- DE-102017103243-B3: Teaches train detector (10) with a holder (20) for attachment to a rail (100) and a sensor (30) received by the holder (20) for detecting a train passing the train detector (10), characterized in that the holder (20) comprises at least one magnet (40) for attaching the train detector (10) to the rail web (104) and a lever extending away from the magnet (40) for releasing the train detector (10) from the rail (100).
- CN-110316225-A: Teaches a track traffic vehicle wheel height comprises the sensor mounting structure, the rail clamp is fixed at the bottom of the steel rail, for mounting the wheel to the sensor protective cover, and a protective cover for adjusting the height of the adjusting support base, the height adjusting bracket is an insulating bracket.
- EP 3567161 A1: Teaches a plug for installation of wired sensor track connectors, including a through body (1) and a plug socket (2) having at least four connectors, wherein the body (1) is provided with a base to be placed in a detachable manner on a centering element (4) configured as an angled profile made of a load-bearing plate and an adjusting plate (6) perpendicular to the load-bearing plate.

25. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CHENG XI LIN whose telephone number is (571)272-6102. The examiner can normally be reached Mon. through Fri. 9:00am to 6:00pm EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Samuel (Joe) Morano can be reached at 5712726684. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/CHENG LIN/
Examiner, Art Unit 3615